QueueLess – Complete Feature Specification

1. Project Overview

QueueLess is a web-based digital queue and token management system that allows customers to join service queues remotely, receive digital tokens, monitor their queue position, and receive service without standing in a physical queue.

The system contains three main panels:

1. **Customer Panel**
2. **Staff Panel**
3. **Admin Panel**

Technology Stack

- Frontend: HTML, CSS, JavaScript, ReactJs,Redux
- Backend: Python + FastAPI
- Database: PostgreSQL
- Authentication: JWT
- Password Security: Argon2/Bcrypt
- API Architecture: REST API

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2. Main System Features

2.1 User Management

The system shall provide user account management for customers, staff, and administrators.

Features:

- User registration
- User login
- User logout
- Password hashing

- Password change
- Profile management
- Profile update
- Account activation/deactivation
- Role-based access
- Authentication token management
- Session/token expiration
- Unauthorized access protection

Supported roles:

- CUSTOMER
- STAFF
- ADMIN

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3. Customer Panel Features

3.1 Customer Registration

Customers can create an account.

Fields:

- Full Name
- Email
- Mobile Number
- Password
- Confirm Password

Validation:

- Required field validation
- Email format validation
- Mobile number validation
- Password strength validation
- Duplicate email prevention
- Duplicate phone prevention
- Password confirmation validation

3.2 Customer Login

Customers can securely log into the system.

Features:

- Email/phone login
- Password authentication
- Remember session where applicable
- Invalid credential message
- Account inactive message
- Automatic redirection to Customer Dashboard

3.3 Customer Dashboard

The customer dashboard provides an overview of the customer's current queue activity.

Dashboard information:

- Welcome message
- Active token
- Current token number
- Current serving token
- People ahead
- Estimated waiting time
- Selected service
- Counter information
- Token status
- Queue status
- Recent token history

Dashboard actions:

- Join Queue
- View Token
- Cancel Token
- View History
- Submit Feedback

3.4 View Available Services

Customers can view all services currently available.

Each service can display:

- Service name
- Service description
- Token prefix
- Current queue length
- Currently serving token
- Estimated waiting time
- Service availability
- Number of active counters

Example:

Driving Licence Service
Current Token: DL-105
People Waiting: 8
Estimated Wait: 32 Minutes

[Join Queue]

3.5 Service Details

Customers can open a service to view detailed queue information.

Information:

- Service name
- Description
- Queue status
- Current token
- Queue length
- Estimated waiting time
- Counter availability
- Operating hours
- Queue closing time
- Join Queue option

3.6 Join Queue

The customer can join an available queue.

Process:

Select Service
↓
Check Queue Availability
↓
Confirm Join
↓
Generate Token
↓
Display Token

System checks:

- Service is active
- Queue is open
- Customer has no active token
- Customer is eligible to join
- Queue capacity rules if configured

3.7 Token Generation

After successfully joining a queue, the system generates a unique token.

Example:

E-128
DL-042
OP-109

Token information:

- Token ID
- Token number
- Token code
- Service
- Date
- Time
- Status

- Position
- Estimated waiting time

The token must be unique for the appropriate queue/date/service configuration.

3.8 Active Token Tracking

Customers can continuously monitor their active token.

Information:

- My token
- Current serving token
- Tokens ahead
- Queue position
- Estimated waiting time
- Service name
- Counter
- Status

Possible states:

WAITING
CALLED
SERVING
COMPLETED
SKIPPED
CANCELLED

3.9 Queue Position

The system displays the customer's current queue position.

Example:

Your Token: E-128

Current Token: E-123
People Ahead: 4
Your Position: 5

The position should update whenever earlier tokens are completed, skipped, or cancelled.

3.10 Estimated Waiting Time

The system calculates an approximate waiting time.

Example:

Estimated Waiting Time:
24 Minutes

Calculation may consider:

- Number of people ahead
- Average service time
- Number of active counters
- Current serving token
- Queue configuration

3.11 Token Status

Customers can see token status.

Possible statuses:

- Waiting
- Called
- Serving
- Completed
- Skipped
- Cancelled

Status changes should be reflected in the customer interface.

3.12 Token Cancellation

Customers can cancel their own waiting token.

Rules:

- Only eligible active tokens can be cancelled
- Completed tokens cannot be cancelled
- Serving tokens cannot be cancelled by customer
- Cancelled token cannot return to the queue

Confirmation:

Are you sure you want to cancel your token?

[Cancel] [Confirm]

3.13 Counter Information

After a token has been called, the customer should be able to see:

- Counter number/name
- Service name
- Token number
- Called status
- Call time

Example:

Token: E-128
Please proceed to:
Counter 03

3.14 Token History

Customers can view previously generated tokens.

History information:

- Token number
- Service
- Date
- Time

- Counter
- Status
- Completion time

Filters:

- Date
- Service
- Status

3.15 Customer Feedback

Customers can submit feedback after service completion.

Fields:

- Rating
- Comments

Rating example:

★ ★ ★ ★ ★

Rules:

- Feedback only after completed service
- One feedback record per eligible token
- Rating required
- Optional comments

3.16 Customer Profile

Customers can:

- View profile
- Edit name
- Update phone
- Update email where permitted
- Change password
- View account status

- Logout

3.17 Customer Notifications

The system can notify customers about important token events.

Possible notifications:

- Token generated
- Queue position changed
- Token called
- Counter assigned
- Service completed
- Queue cancelled/closed

Notification channels can later include:

- Web notification
- Email
- SMS
- Browser notification

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4. Staff Panel Features

4.1 Staff Login

Staff members can securely log into the Staff Panel.

Validation:

- Staff credentials
- Account status
- Role validation
- Assigned counter validation

4.2 Staff Dashboard

The dashboard provides the current operational status.

Dashboard information:

- Staff name
- Assigned counter
- Assigned service
- Current token
- Waiting tokens
- Completed tokens
- Skipped tokens
- Today's total customers
- Average service time

4.3 Assigned Counter

Staff can view their assigned counter.

Information:

- Counter name/number
- Service
- Counter status
- Staff assignment
- Operating status

Example:

Counter: C-03
Service: Examination
Status: OPEN
Staff: John

4.4 Waiting Queue

Staff can view customers currently waiting.

Information:

- Token number
- Queue position

- Created time
- Waiting duration
- Status

Example:

E-125 Waiting
E-126 Waiting
E-127 Waiting
E-128 Waiting

4.5 Call Next Token

Staff can call the next customer.

Process:

Waiting Queue
↓
Call Next
↓
Token = CALLED
↓
Display Token

The system should select the appropriate next eligible token.

4.6 Current Token

Staff can view the token currently being handled.

Information:

- Token number
- Customer name/reference
- Service
- Call time
- Start time
- Current status

4.7 Start Service

Staff can start serving a called customer.

Status transition:

CALLED → SERVING

System records:

- Start time
- Staff
- Counter
- Token status

4.8 Complete Service

After service is finished, staff can mark the token as completed.

Status transition:

SERVING → COMPLETED

System records:

- Completion time
- Staff
- Counter
- Service duration

4.9 Skip Token

Staff can skip a customer when required.

Status transition:

CALLED → SKIPPED

Possible reason:

- Customer unavailable
- Customer did not respond
- Customer left queue

The system may optionally require a skip reason.

4.10 Recall Token

Staff can recall a previously called token.

Use case:

Token E-128
↓
Customer did not hear
↓
Recall
↓
Token E-128 called again

The token remains eligible according to the configured rules.

4.11 Queue Pause/Resume

Staff may be allowed to pause queue processing depending on system configuration.

Examples:

- Temporary counter break
- Technical issue
- Emergency
- Lunch break

Possible states:

OPEN
PAUSED
CLOSED

4.12 Staff Statistics

Staff can view daily operational statistics.

Statistics:

- Total tokens handled
- Completed tokens
- Skipped tokens
- Cancelled tokens
- Current waiting count
- Average service time
- Average waiting time
- Tokens processed today

4.13 Staff History

Staff can see tokens handled by their counter.

Information:

- Token number
- Service
- Customer reference
- Start time
- Completion time
- Status
- Service duration

4.14 Staff Profile

Staff can:

- View profile

- Edit allowed information
- Change password
- View assigned counter
- View assigned service
- Logout

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5. Admin Panel Features

5.1 Admin Login

Administrator can securely access the administration panel.

5.2 Admin Dashboard

The Admin Dashboard provides complete system monitoring.

Dashboard statistics:

- Total customers
- Total staff
- Total services
- Total counters
- Active queues
- Waiting customers
- Currently serving tokens
- Completed tokens today
- Cancelled tokens today
- Skipped tokens today

5.3 Customer Management

Admin can manage customer accounts.

Features:

- View customers

- Search customers
- Filter customers
- View customer details
- Activate customer
- Deactivate customer
- Edit customer information
- View customer token history

Search fields:

- Name
- Email
- Phone

5.4 Staff Management

Admin can manage staff members.

Features:

- Add staff
- View staff
- Edit staff
- Activate/deactivate staff
- Reset password
- Assign service
- Assign counter
- Remove assignment
- View staff activity

5.5 Service Management

Admin can create and manage services.

Service fields:

- Service name
- Description
- Token prefix
- Average service time

- Operating status
- Queue capacity
- Start time
- End time

Features:

- Add service
- Edit service
- Activate service
- Deactivate service
- Delete service where permitted
- View queue status

5.6 Counter Management

Admin can manage service counters.

Features:

- Add counter
- Edit counter
- Activate counter
- Deactivate counter
- Assign service
- View assigned staff
- View counter status

Example:

Counter 01 → Passport
Counter 02 → Passport
Counter 03 → Driving Licence

5.7 Staff-Counter Assignment

Admin can assign staff members to counters.

Features:

- Assign staff

- Change counter
- Remove assignment
- Set assignment active/inactive
- View assignment history

Rules:

- Only valid staff can be assigned
- Inactive staff cannot operate
- Assignment conflicts should be prevented according to system rules

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5.8 Queue Monitoring

Admin can monitor all queues.

Information:

- Service
- Current token
- Waiting tokens
- Serving token
- Available counters
- Waiting time
- Queue status

Example:

```
Service       Current   Waiting   Counters
-----------------------------------------------
Passport      P-102     12        3
Licence       L-050     5         2
Examination   E-090     18        4
```

5.9 Queue Controls

Admin may control queue operations.

Features:

- Open queue
- Pause queue

- Resume queue
- Close queue
- Configure queue limits
- Configure token rules

5.10 Queue Settings

Admin can configure queue behavior.

Settings may include:

- Token prefix
- Daily token reset
- Maximum queue size
- Average service time
- Queue opening time
- Queue closing time
- Automatic queue closure
- Waiting time calculation method
- Token cancellation policy

5.11 Token Management

Admin can monitor token records.

Features:

- View all tokens
- Search token
- Filter token
- View token details
- View status
- View customer
- View service
- View counter
- View timestamps

Filters:

- Date

- Service
- Counter
- Status
- Staff

5.12 Token History / Audit Trail

Every important token state change should be recorded.

Example:

E-128

WAITING
↓
CALLED
↓
SERVING
↓
COMPLETED

History should record:

- Token ID
- Old status
- New status
- Changed by
- Date/time
- Staff/counter where applicable

5.13 Reports

Admin can generate reports.

Possible reports:

Daily Report

- Total tokens

- Completed tokens
- Cancelled tokens
- Skipped tokens
- Average waiting time
- Average service time

Service Report

- Tokens per service
- Completed tokens
- Average waiting time
- Average service time

Staff Report

- Tokens handled
- Completion count
- Average service duration

Counter Report

- Counter utilization
- Tokens processed
- Average service time

Customer Report

- Total customers
- Active customers
- Customers served

5.14 Report Filtering

Reports can be filtered by:

- Date
- Date range
- Service
- Counter
- Staff
- Token status

5.15 Dashboard Charts

Admin dashboard can display charts such as:

- Tokens per day
- Tokens per service
- Completed vs cancelled
- Average waiting time
- Average service time
- Counter performance

5.16 Admin Settings

Admin can configure system-level settings.

Examples:

- Application name
- Queue rules
- Token rules
- Operating hours
- Notification settings
- Security settings
- System status

5.17 Admin Profile

Admin can:

- View profile
- Edit profile
- Change password
- Logout

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6. Common System Features

6.1 Authentication

All protected panels must require authentication.

Flow:

Login
↓
Validate Credentials
↓
Generate JWT
↓
Store Token
↓
Access Protected APIs

6.2 Role-Based Authorization

The backend must enforce roles.

Example:

/customer/* → CUSTOMER
/staff/* → STAFF
/admin/* → ADMIN

Frontend button hiding alone must not be treated as security.

6.3 Search

Search should be available where useful.

Search examples:

- Customer name
- Email

- Phone
- Token number
- Service name
- Staff name
- Counter name

6.4 Filtering

Lists should support filtering.

Example:

Status: [All]
Service: [All]
Date: [Today]

6.5 Sorting

Tables can support sorting by:

- Date
- Token
- Status
- Name
- Service
- Waiting time

6.6 Pagination

Large records should use pagination.

Examples:

- Customers
- Staff
- Tokens
- Token history
- Reports

6.7 Notifications / Toast Messages

The frontend should provide clear feedback.

Examples:

Token generated successfully.
Service updated successfully.
Unable to join queue.
Token cancelled successfully.
Invalid login credentials.

6.8 Loading States

Every asynchronous operation should provide a loading state.

Examples:

Joining Queue...
Loading Queue...
Saving...
Generating Report...

6.9 Error Handling

The system should handle:

- Invalid input
- Network failures
- Server errors
- Database errors
- Unauthorized requests
- Expired JWT
- Duplicate records
- Queue unavailable
- Invalid token status

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7. Queue Management Features

7.1 Queue Lifecycle

QUEUE CREATED
↓
QUEUE OPEN
↓
CUSTOMERS JOIN
↓
TOKENS GENERATED
↓
STAFF CALL TOKEN
↓
SERVICE STARTED
↓
SERVICE COMPLETED
↓
NEXT TOKEN

7.2 Token Lifecycle

Primary flow:

WAITING
↓
CALLED
↓
SERVING
↓
COMPLETED

Alternative flows:

WAITING → CANCELLED

CALLED → SKIPPED

CALLED → RECALLED

7.3 Multiple Counters

A service can have multiple counters.

Example:

Service: Passport

Counter 1 → Staff A
Counter 2 → Staff B
Counter 3 → Staff C

The queue system should distribute customers among available counters according to configured rules.

7.4 Current Serving Token

Each service should display the token currently being served.

Example:

Now Serving: P-104
Counter: 02

7.5 Queue Position Calculation

The system should calculate a customer's position based on eligible tokens ahead of them.

7.6 Waiting Time Calculation

Estimated waiting time can be derived from:

People Ahead × Average Service Time

For multiple counters, the calculation can be adjusted using active counter count.

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8. Business Rule Features

The system must enforce the following rules.

BR-01

A customer cannot have more than one active token at the same time.

BR-02

A customer can only join an active/open service.

BR-03

A token must belong to a valid customer and service.

BR-04

Token numbers must not duplicate within the defined queue scope.

BR-05

Token generation must be transaction-safe.

BR-06

Completed tokens cannot be cancelled.

BR-07

Cancelled tokens cannot be called.

BR-08

Skipped tokens cannot be treated as waiting tokens unless explicitly re-queued by an authorized rule.

BR-09

Only assigned staff can operate a counter.

BR-10

Staff cannot simultaneously serve multiple tokens at the same counter.

BR-11

A staff member cannot call another token while actively serving another token on the same counter.

BR-12

Only authorized roles can perform role-specific operations.

BR-13

Inactive services cannot accept new customers.

BR-14

Inactive counters cannot process tokens.

BR-15

Inactive staff cannot operate counters.

BR-16

Token numbering can reset according to configured queue rules, normally per day/service.

BR-17

Every important token status change should be logged.

BR-18

Customers should only be allowed to cancel their own eligible tokens.

BR-19

Customers should only be allowed to view their own private token information.

BR-20

Feedback should be connected to a valid completed service/token.

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9. Security Features

Authentication Security

- Password hashing
- JWT authentication
- Token expiration
- Secure password change
- Protected endpoints
- Role verification

Authorization Security

- Customer API protection
- Staff API protection
- Admin API protection
- Resource ownership validation

Input Security

- Request validation
- SQL injection protection through ORM/parameterized queries
- Invalid input rejection
- Server-side validation

API Security

- Proper HTTP status codes
- Authentication middleware
- Authorization middleware
- Error handling without exposing sensitive information

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10. UI/UX Features

Common UI Components

- Navbar
- Sidebar
- Footer
- Dashboard cards
- Buttons
- Forms
- Input fields
- Select boxes
- Tables
- Cards
- Modals
- Confirmation dialogs
- Toast notifications
- Alerts
- Badges
- Pagination
- Search
- Filters
- Loading indicators
- Empty states

Customer UI Components

- Service Card
- Token Card
- Queue Status Card
- Waiting Time Card
- Current Token Display
- Token History Table
- Feedback Form
- Cancellation Modal

Staff UI Components

- Current Token Card
- Waiting Queue List
- Queue Item
- Call Button
- Start Button
- Complete Button
- Skip Button
- Recall Button
- Counter Status Card
- Statistics Cards

Admin UI Components

- Statistics Cards
- Customer Table
- Staff Table
- Service Table
- Counter Table
- Token Table
- Search Box
- Filters
- Add/Edit Modal
- Confirmation Modal
- Status Badge
- Charts
- Reports

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11. UI State Features

All important operations should support the following states:

Normal

[Join Queue]

Loading

[Joining...]

Success

Token Generated Successfully
Your Token: E-128

Error

Unable to join the queue.
Please try again.

Disabled

Queue Closed
[Join Queue]

Empty

No customers are currently waiting.

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12. Optional Future Features

These features can be added after the core system is completed.

QR Token

Customer receives a QR code containing token information.

Digital Display

A public display can show:

NOW SERVING

E-127 → Counter 02
E-128 → Counter 01

Real-Time Queue Updates

Use WebSocket technology so token changes are reflected without page refresh.

SMS Notifications

Send token-call notifications through SMS.

Email Notifications

Send token and queue updates through email.

Mobile Responsive Design

Optimize all panels for:

- Desktop
- Tablet
- Mobile

Appointment Booking

Allow customers to reserve a future service slot.

Priority Queue

Support categories such as:

- General
- Senior Citizen
- Emergency
- Priority

QR Check-In

Customer scans a QR code to join a queue.

Advanced Analytics

- Peak hours
- Service demand
- Counter utilization

- Average waiting trends
- Staff workload analysis

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13. Core MVP Features

For the first working version, the essential features should be:

Admin

- Admin login
- Service creation
- Counter creation
- Staff creation
- Staff-counter assignment
- Queue monitoring

Customer

- Registration
- Login
- View services
- Join queue
- Generate token
- View active token
- View queue position
- Cancel token
- View token status

Staff

- Login
- View assigned counter
- View waiting queue
- Call next token
- Start service
- Complete service
- Skip token
- Recall token

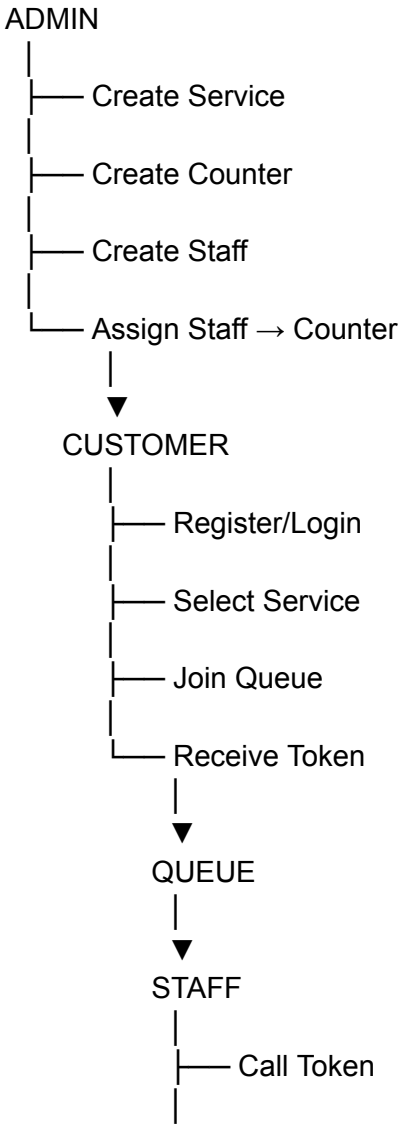
System

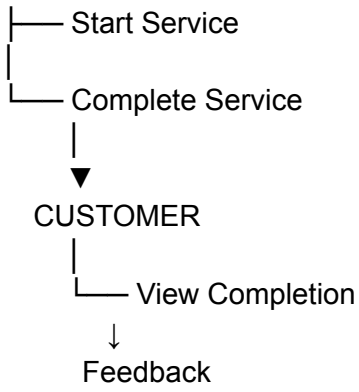
- JWT authentication
- Role-based authorization
- PostgreSQL database
- Transaction-safe token generation
- Token status management
- Token history
- Basic error handling

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14. Final Core Workflow

The minimum complete business workflow is:





15. Feature Priority

Priority 1 – Mandatory

- Authentication
- Role management
- Customer registration/login
- Staff login
- Admin login
- Service management
- Counter management
- Staff management
- Staff assignment
- Queue creation/opening
- Token generation
- Queue position
- Call next
- Start service
- Complete service
- Skip token
- Token status
- Token history
- Role-based authorization

Priority 2 – Important

- Waiting time calculation
- Token cancellation
- Recall token

- Customer history
- Staff statistics
- Admin queue monitoring
- Reports
- Feedback
- Search
- Filtering
- Pagination

Priority 3 – Future Enhancement

- WebSocket real-time updates
- QR code
- SMS
- Email notifications
- Digital display
- Priority queue
- Appointment booking
- Advanced analytics
- Mobile optimization

16. Core QueueLess Goal

The complete system should successfully support this scenario:

Admin creates a service
↓
Admin creates a counter
↓
Admin creates staff
↓
Admin assigns staff to counter
↓
Customer logs in
↓
Customer selects service
↓
Customer joins queue
↓
System generates unique token
↓
Customer tracks queue position

↓
Staff calls token
↓
Customer visits counter
↓
Staff starts service
↓
Staff completes service
↓
Token becomes COMPLETED
↓
Customer views completed token
↓
Customer submits feedback

This workflow represents the **core business functionality of QueueLess**.